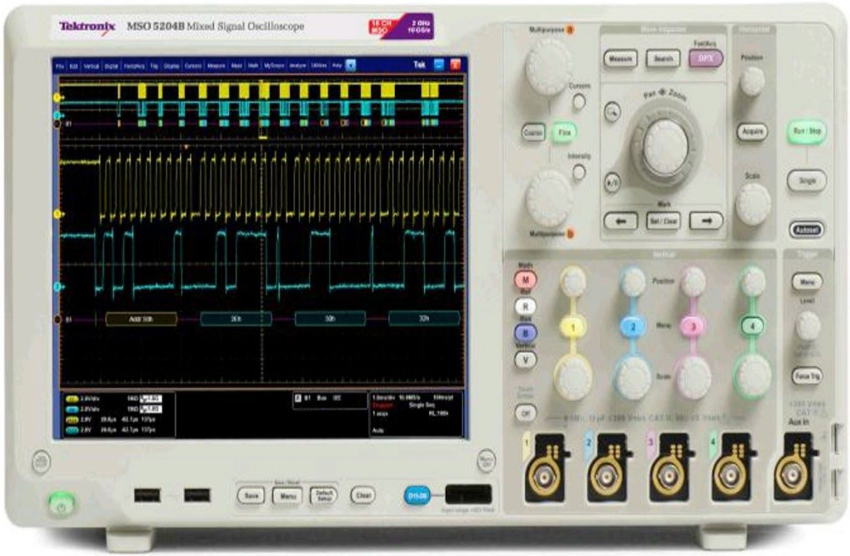
Signal measurement

CBD_SOLO_PE
09/12/2023
Version 01

I. Measure Voltage and Signal

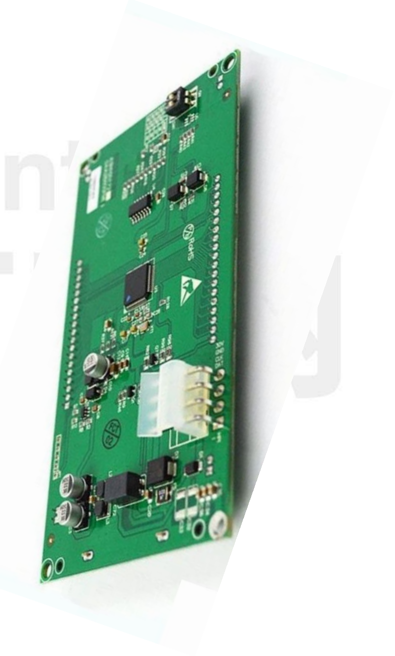
Prepare

1. Oscilloscope
2. Probe for measure voltage
3. Device



Oscilloscope

Voltage probe



Device

A. Measure Voltage

A1. Setup

Step 1: Determine the location to be measured

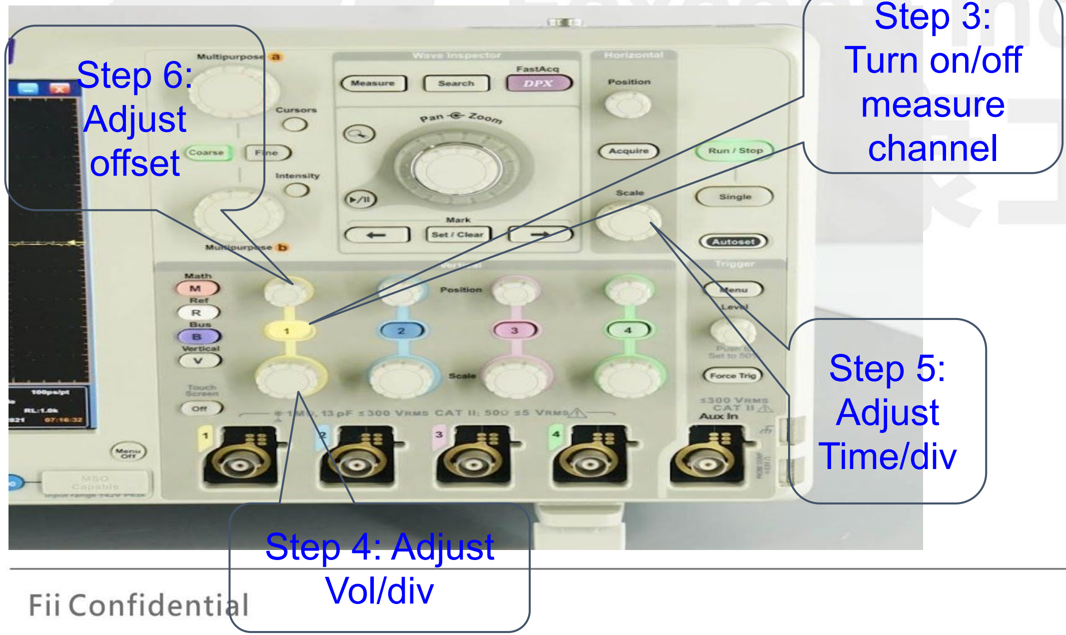
Step 2: Power on device need to measure

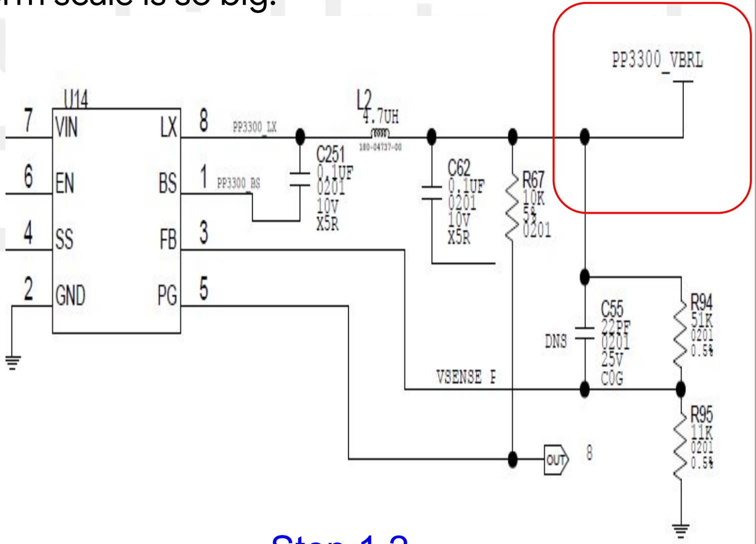
Step 3: Turn on channel to measure (ex: Channel 1 by press button “1” on the Oscilloscope)

Step 4: Adjust Vol/div by knob as bellow picture. Vol/div is to adjust the size of the voltage scale

Step 5: Adjust Time/div by knob as bellow picture. Time/Div is to adjust the length of time scale

Step 6 (option): Adjust offset by knob as bellow picture. This function is to adjust the waveform displayed in the frame of the measurement screen. This function will be use if the waveform scale is so big.





Step 1,2

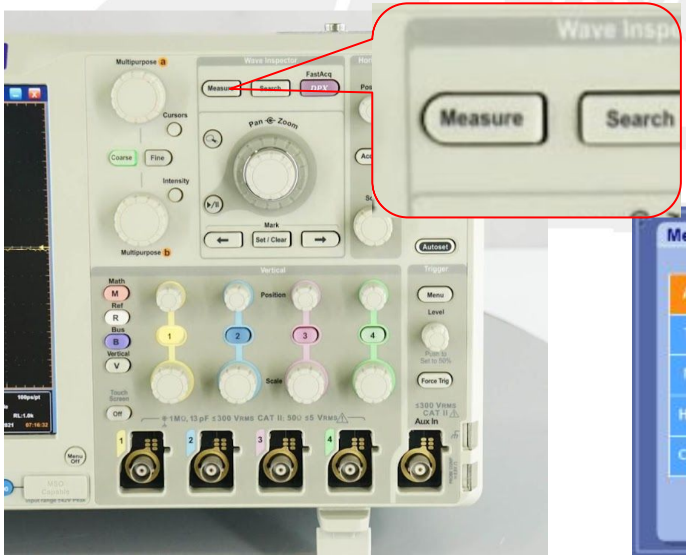
A1. Setup

Step 7: On the machine press “Measure” to show the “Measurement Setup” table

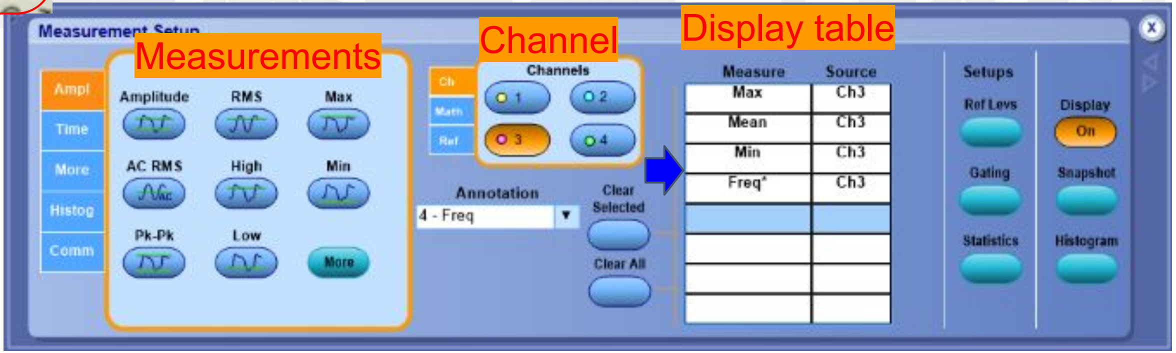
- Select the “Channel” will be used to measure
- Select the item to use in the “Measurements” table

And then the item that select before will be show on the “Display table”

Ex: Select Max, Min, Mean, Pk-Pk. These data will show directly on the measurement screen as below.



		Value	Mean	Min	Max	St Dev	Count	Info
C2	Max	3.52V	3.4948312	3.48	3.56	19.63m	187.0	
C2	Min	3.16V	3.1558532	3.08	3.16	12.33m	187.0	
C2	Mean	3.319V	3.3191231	3.316	3.323	1.358m	187.0	
C2	Pk-Pk*	360.0mV	338.97798m	320.0m	400.0m	22.7m	187.0	

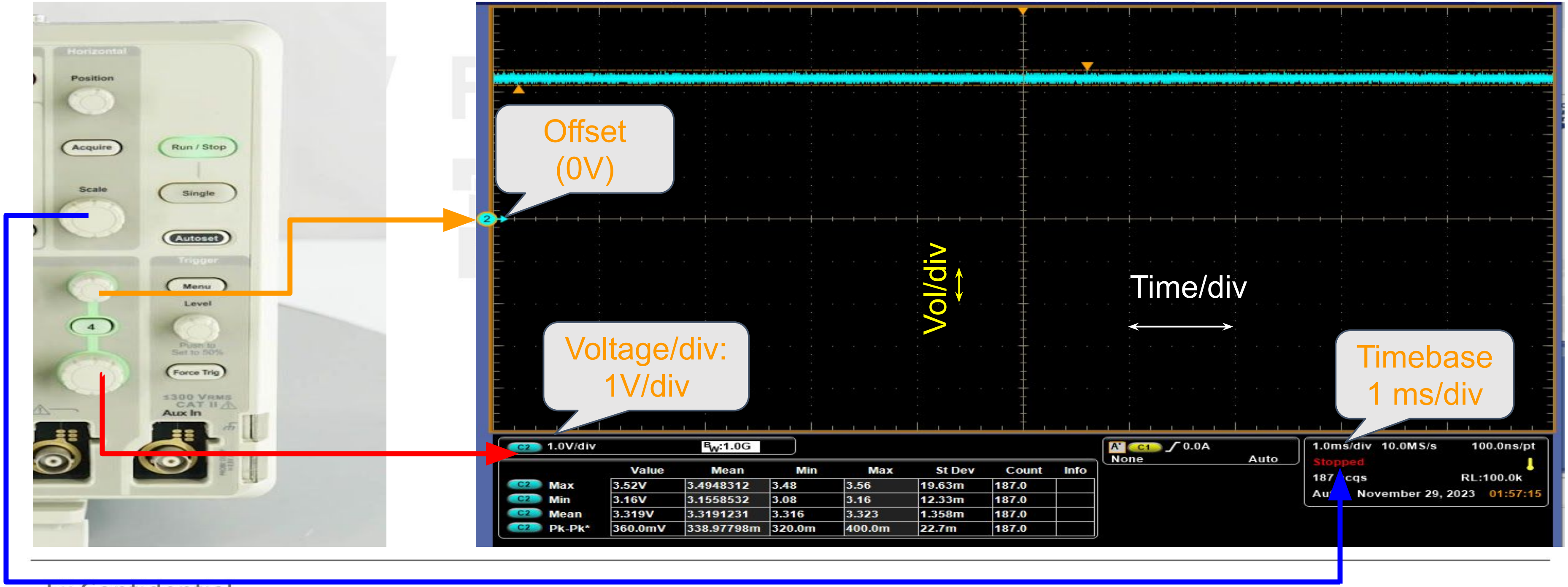


A2. Measure

Measure 3.3V power

- Select Vol/div suitable with Voltage measured (ex: Select 1V/div for measure power 3V3)
- Select Timebase: If wants to see details chose Timebase small ex 10ns/div

  If wants to see the overview chose Timebase large ex 1ms/div



B. Measure Ripple Voltage

B1. Setup

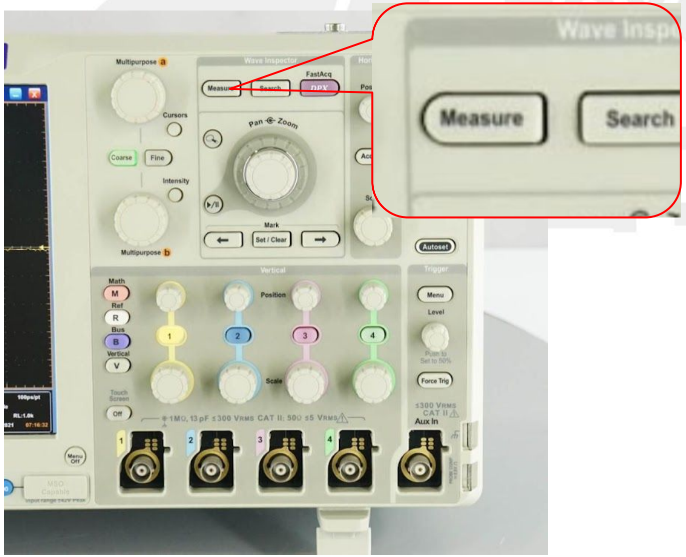
Step 1 →6: Same as slide Measure voltage

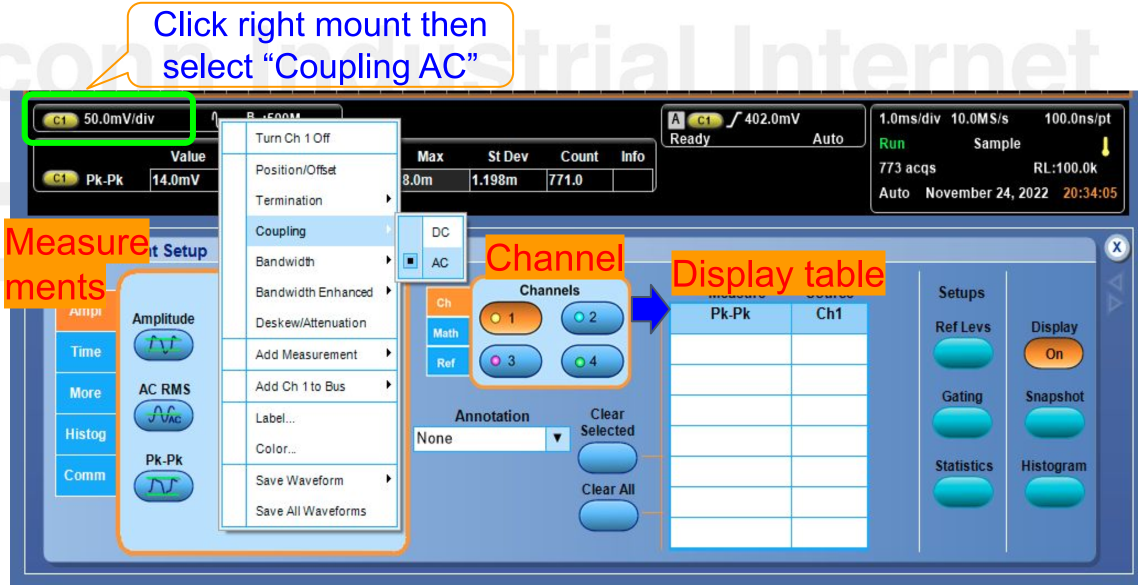
Step 7: On the machine press “Measure” to show the “Measurement Setup” table

The item that select before will be show on the “Display table”

Ex: Select Pk-Pk. These data will show directly on the measurement screen as below.

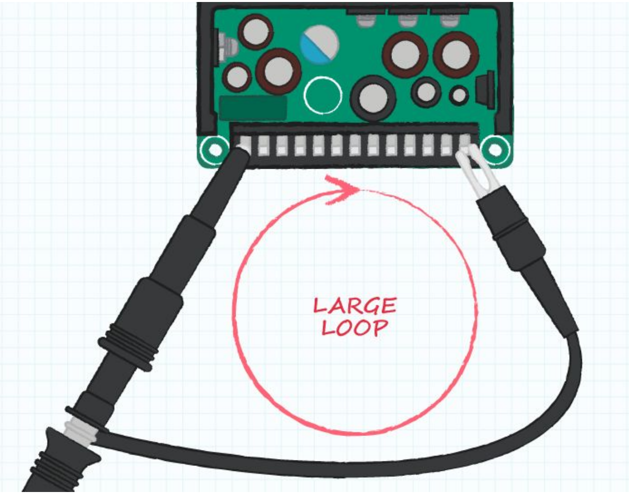
Step 8: Click right mount on “C1” then select Coupling to AC for measure ripple voltage

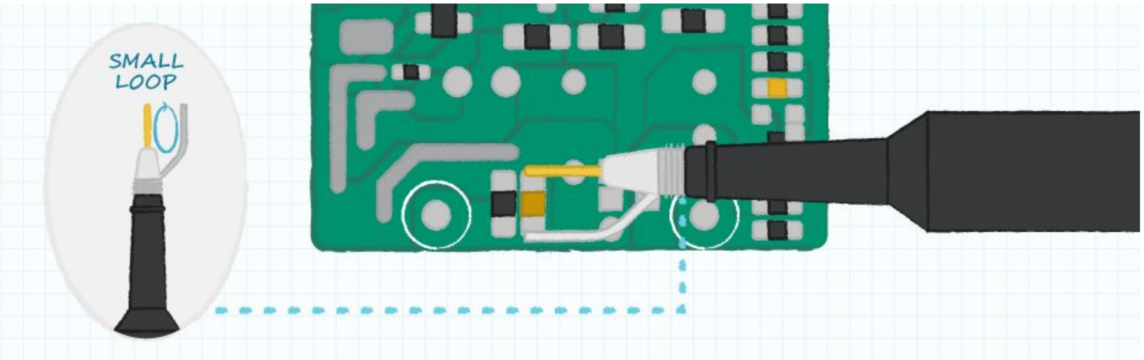




B1. Setup

Step 9: Checking and make a smallest loop as can





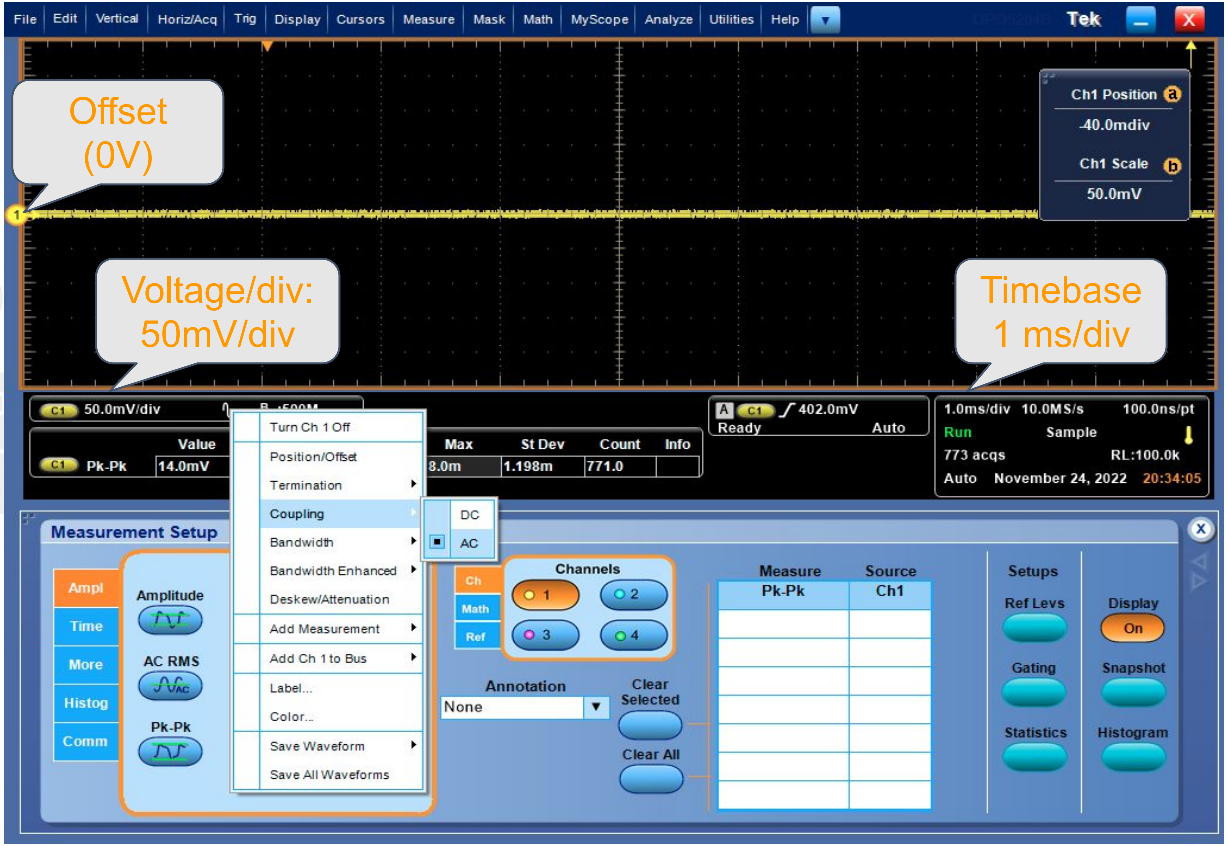
B2. Measure

Measure Ripple of 3.3V power:

- Select Vol/div suitable with Ripple Voltage measured (ex: Select 50mV/div for measure power 3V3)

- Select Timebase:

+ If wants to see details chose Timebase small (ex 10ns/div).
+ If wants to see the overview chose Timebase large (ex 1ms/div).



C. Measure Power Sequence

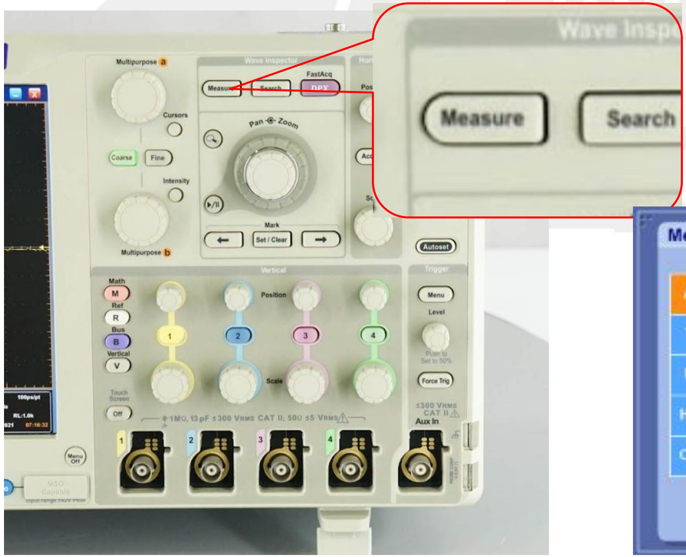
C1. Setup

Step 1 →6: Same as slide Measure voltage

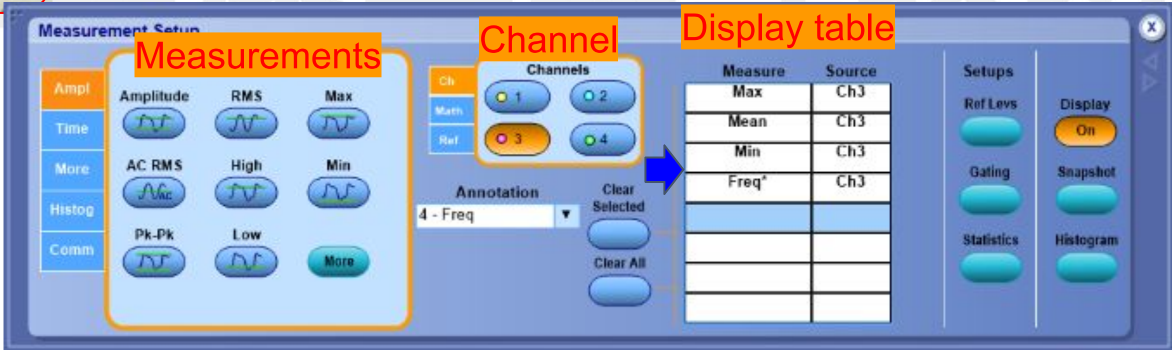
Step 7 (Option): On the machine press "Measure" to show the "Measurement Setup" table

The item that select before will be show on the "Display table"

Ex: Select Max, Min, Mean. These data will show directly on the measurement screen as below.

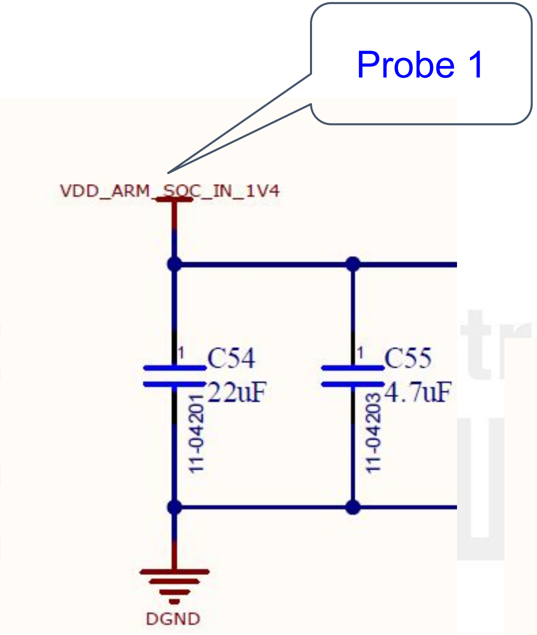


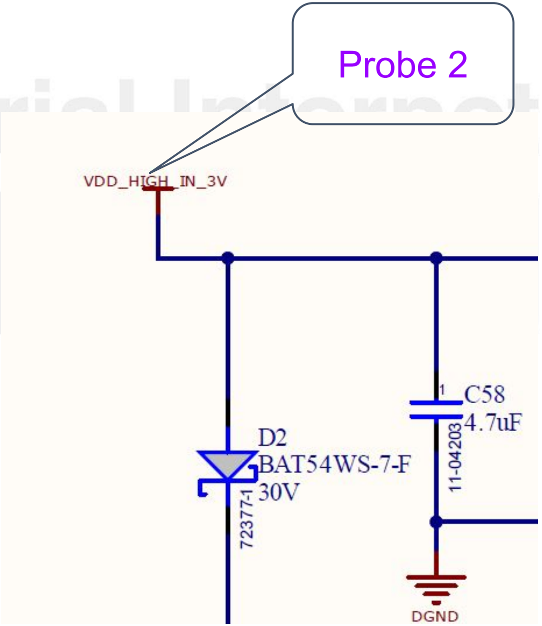
		Value	Mean	Min	Max	St Dev	Count	Info
C2	Max	3.43V	3.4299999	3.43	3.43	0.0	1.0	
C2	Min	-182.0mV	-182.0m	-182.0m	-182.0m	0.0	1.0	
C2	Mean	2.591V	2.5906111	2.591	2.591	0.0	1.0	
C3	Max	1.582V	1.5819999	1.582	1.582	0.0	1.0	
C3	Min	-98.0mV	-98.000119m	-98.0m	-98.0m	0.0	1.0	
C3	Mean*	450.7mV	450.69759m	450.7m	450.7m	0.0	1.0	



C1. Setup

Step 8: Setup the probe on the location on board need measured





4.2.1 Power-Up Sequence

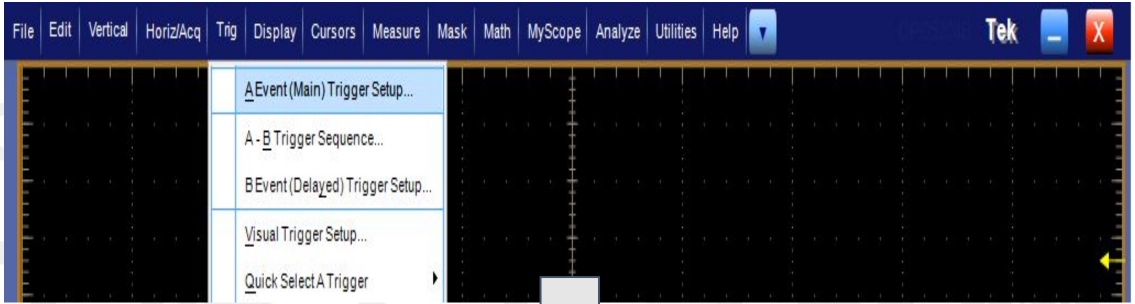
The below restrictions must be followed:

- VDD_SNVS_IN supply must be turned on before any other power supply.
- If a coin cell is used to power VDD_SNVS_IN, then ensure that it is connected before any other supply is switched on.
- VDD_HIGH_IN should be turned on before VDD_SOC_IN.

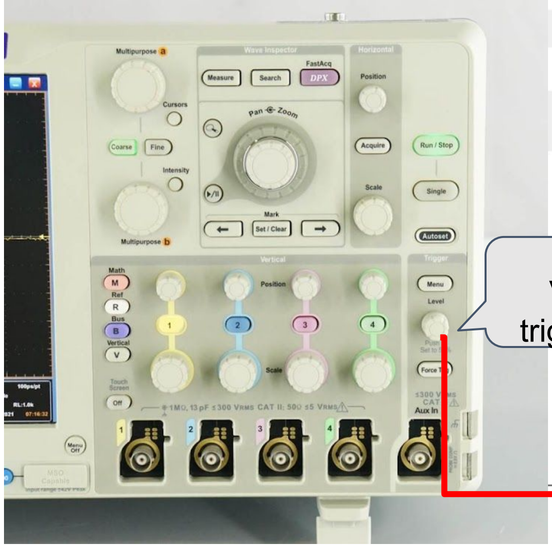
C1. Setup

Step 9: Setup the trigger
(If a pulse appears that reaches the set point, it will enter the interrupt state)

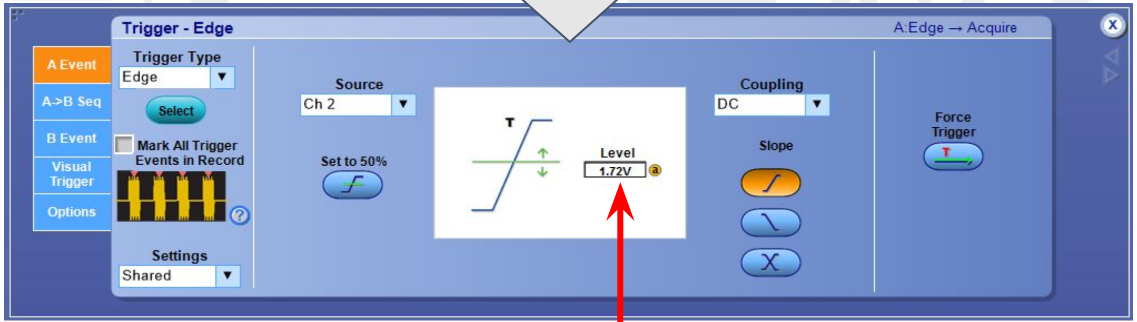
In screen select Trig / A Event (Main) Trigger Setup. → Will show Trigger-Edge table:
+ In Trigger Type select Edge
+ In Slope select Rising edge pulse
+ In Source select Measurement Channel



Setup trigger about 50% of Voltage need measured



Adjust Voltage trigger level

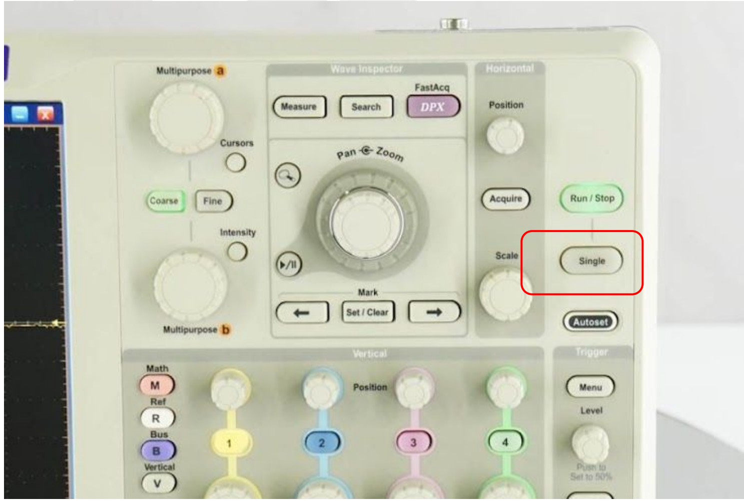


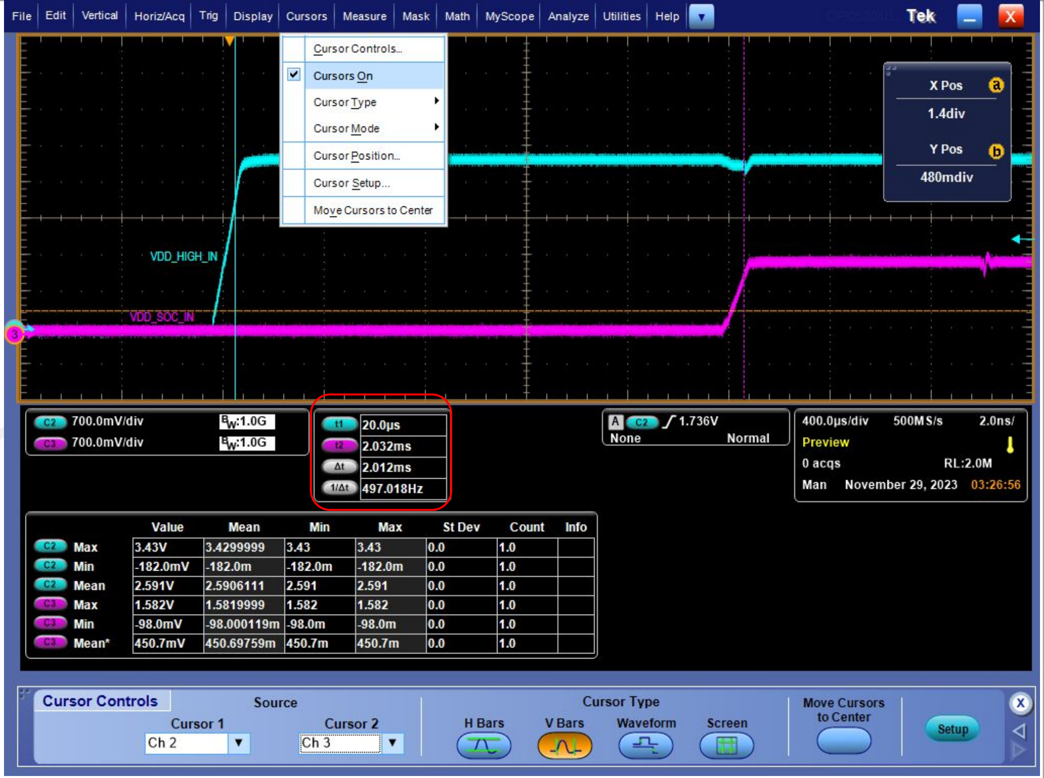
C2. Measure

Measure the Power-Up Sequence of VDD_HIGH_IN and VDD_SOC_IN:

Step 10: In Oscilloscope choose "Single"
Step 11: Turn power of board on"

After turn on power, the oscilloscope will detect and show the signal on the display





4.2.1 Power-Up Sequence

The below restrictions must be followed:

- VDD_SNVS_IN supply must be turned on before any other power supply.
- If a coin cell is used to power VDD_SNVS_IN, then ensure that it is connected before any other supply is switched on.
- VDD_HIGH_IN should be turned on before VDD_SOC_IN.

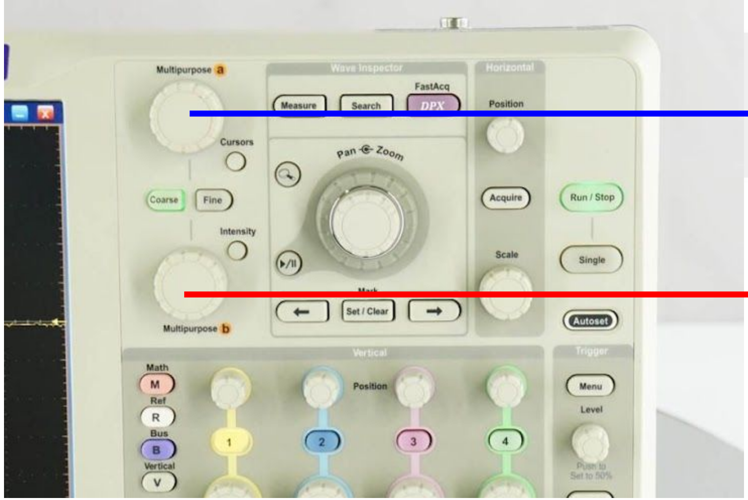
C2. Measure

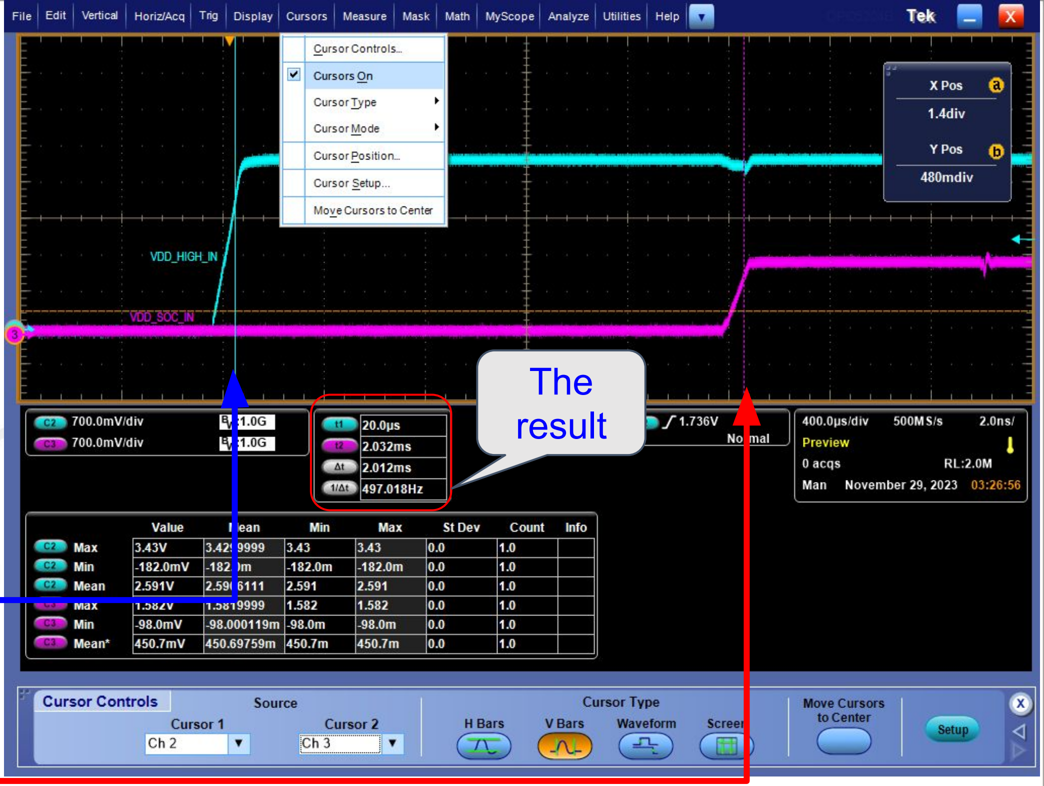
Measure the Power-Up Sequence of VDD_HIGH_IN and VDD_SOC_IN:

Step 12: On display Chose Cursors → Cursors On then select VBars

Step 13: Adjust the bar (by button on oscilloscope) about 80% V measure

→ The result will show at **Δt**





4.2.1 Power-Up Sequence

The below restrictions must be followed:

- VDD_SNVS_IN supply must be turned on before any other power supply.
- If a coin cell is used to power VDD_SNVS_IN, then ensure that it is connected before any other supply is switched on.
- VDD_HIGH_IN should be turned on before VDD_SOC_IN.

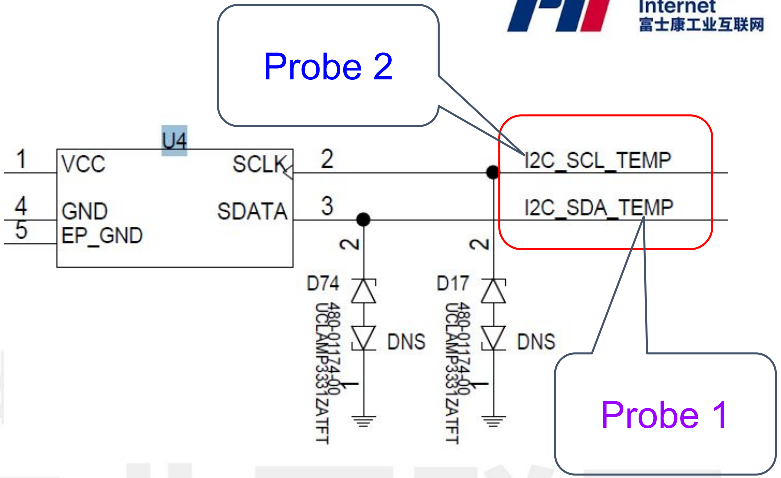
D. Measure Signal

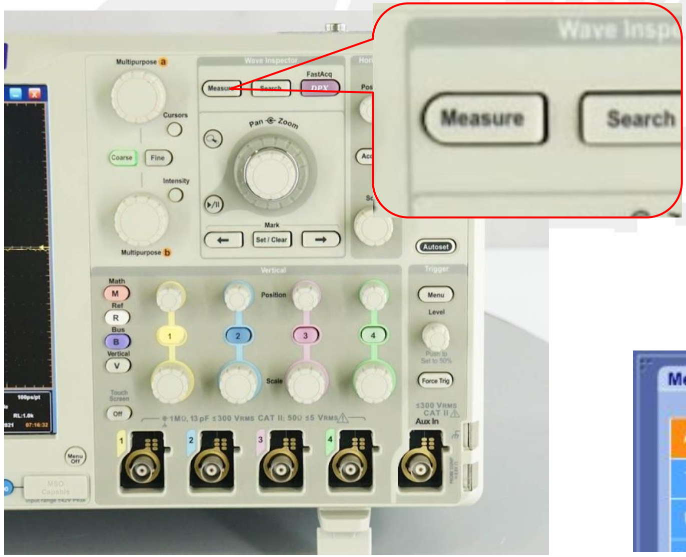
D1.Setup

Step 1 →7: Same as slide Measure voltage

Step 8: Setup the probe on the location on board need measured

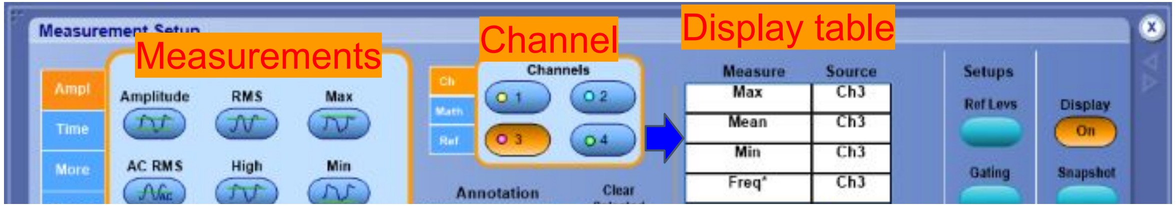
Step 9: Setup trigger level about 50% of Voltage need measured





C1 1.0V/div BW:500M
C2 1.0V/div BW:1.0G

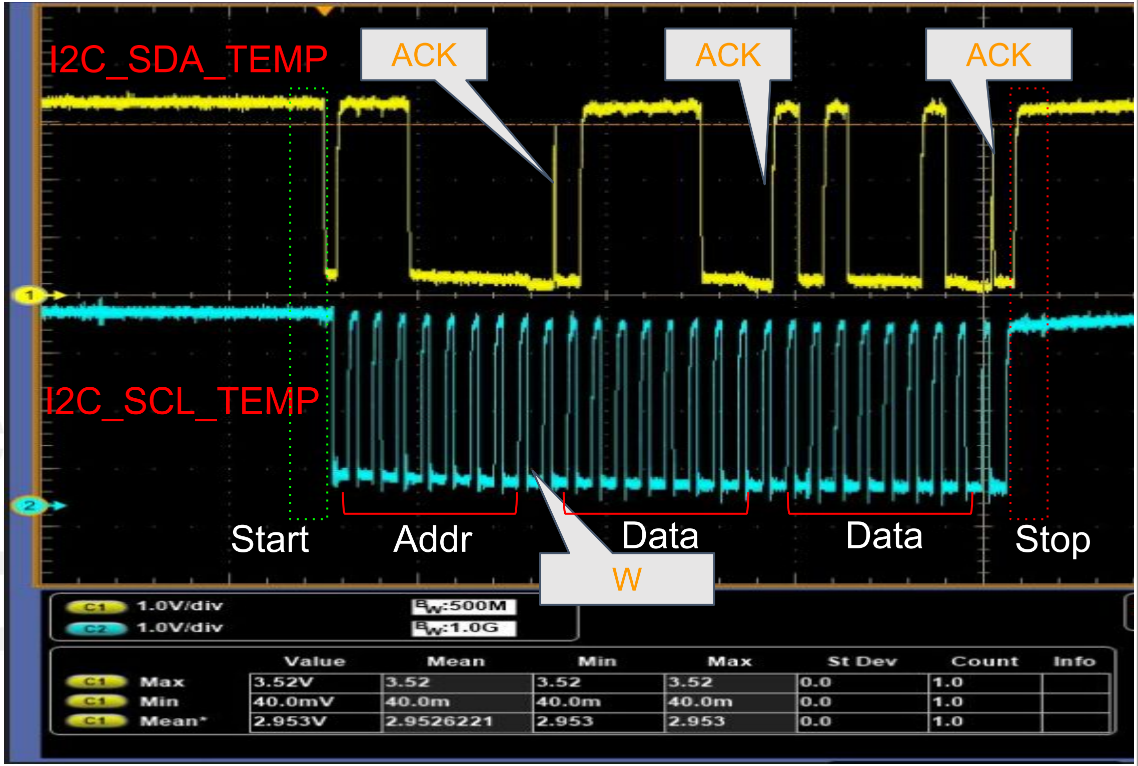
	Value	Mean	Min	Max	St Dev	Count	Info
C1 Max	3.52V	3.52	3.52	3.52	0.0	1.0	
C1 Min	40.0mV	40.0m	40.0m	40.0m	0.0	1.0	
C1 Mean*	2.953V	2.9526221	2.953	2.953	0.0	1.0	

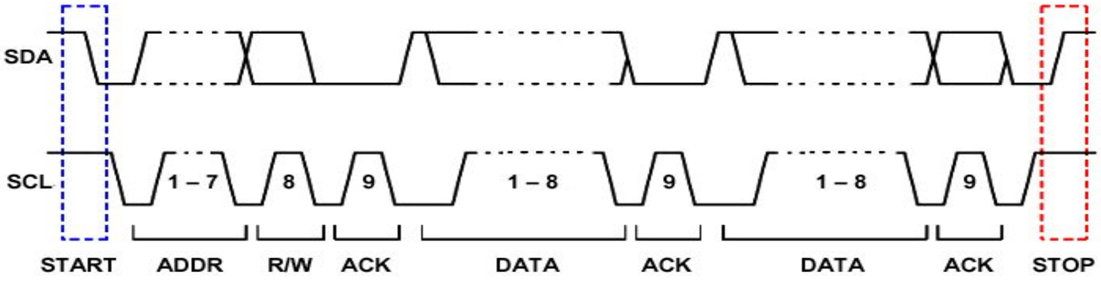


D2. Measure

Step 10: In Oscilloscope choose "Single"
Step 11: Send command to use I2C

→ The result will show as picture





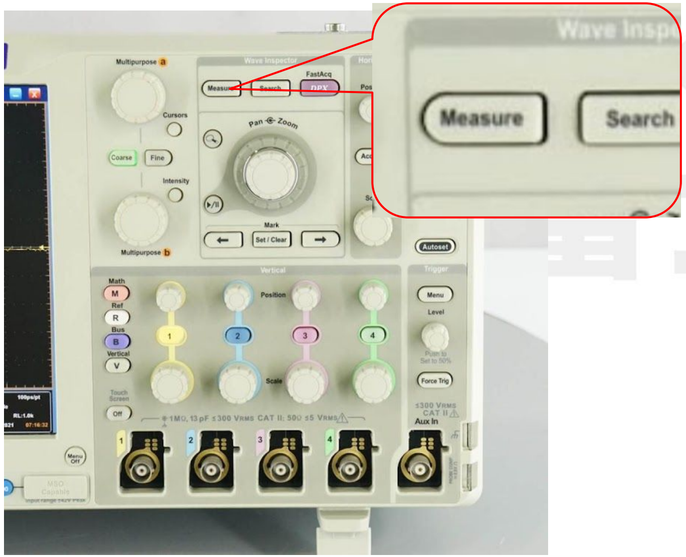
E. Measure Oscillation

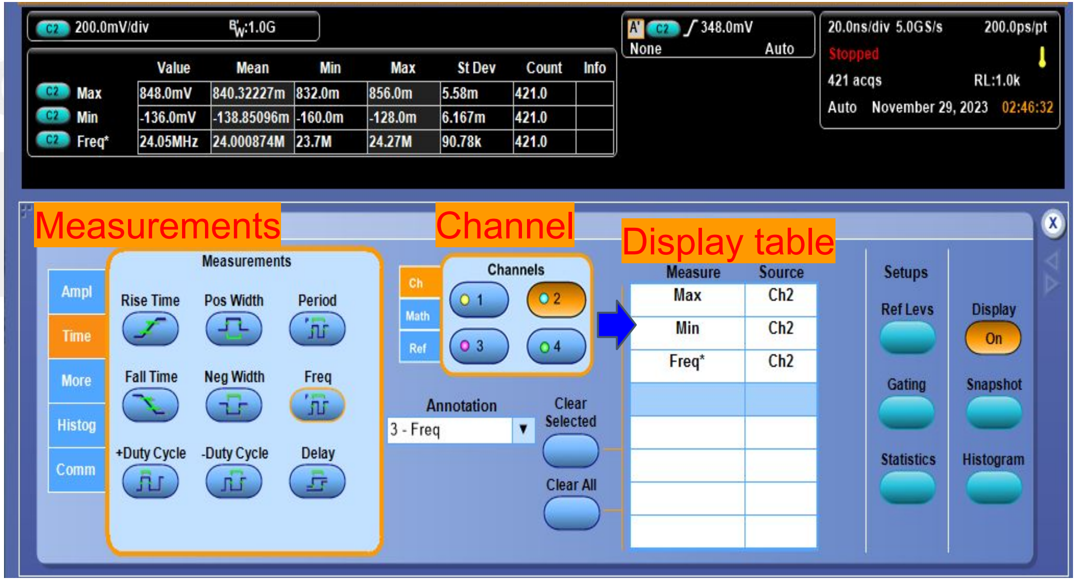
E1. Setup

Step 1: On the machine press “Measure” to show the “Measurement Setup” table

The item that select before will be show on the “Display table”

Ex: Select Max, Min, Freq. These data will show directly on the measurement screen as below.



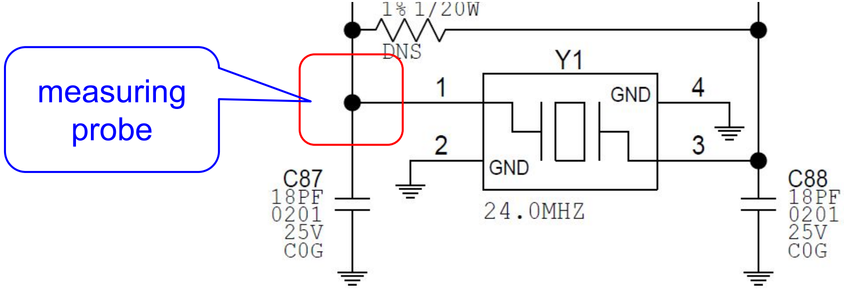


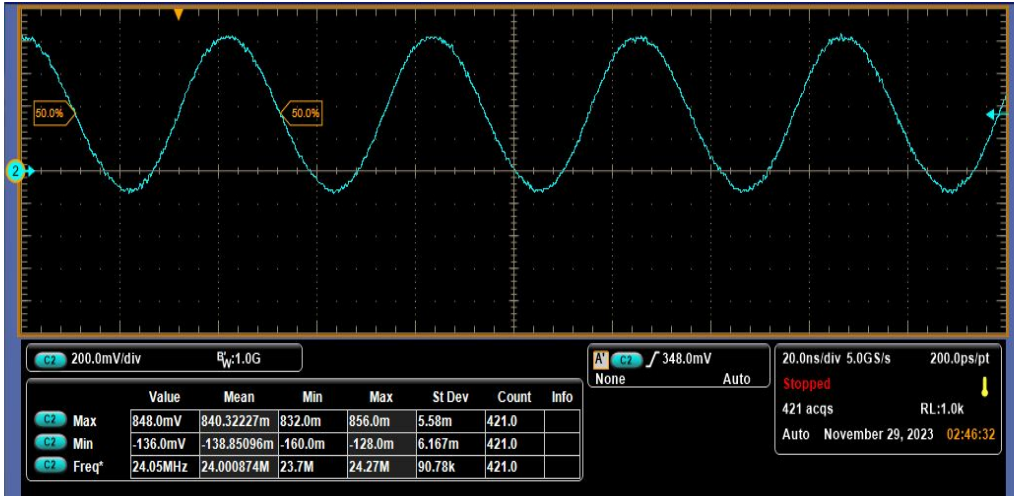
E2. Measure

Measure XTAL 24MHz:

Step 2: In oscilloscope choose “Autoset”

→ The oscilloscope will be autoset and show the waveform of the XTAL signal

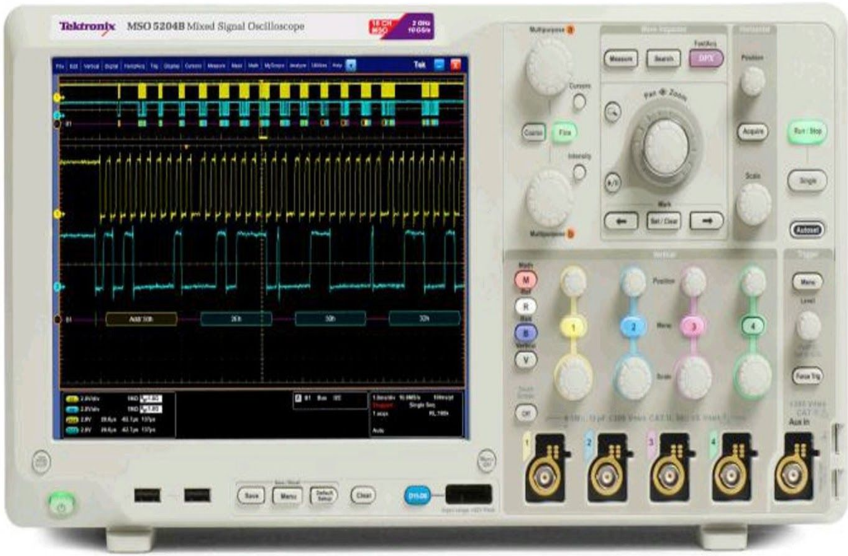




II. Measure Current and Inrush current

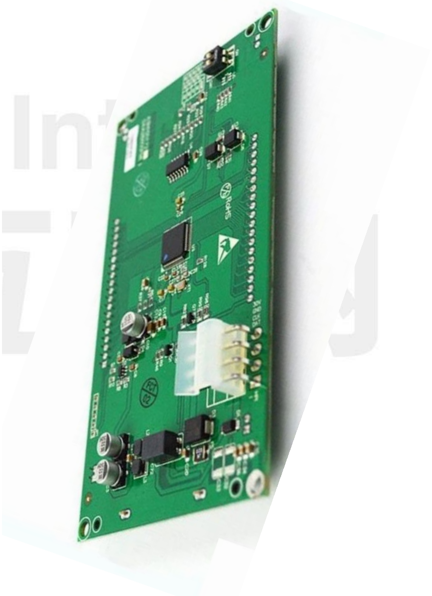
Prepare

1. Oscilloscope
2. Probe for measure current
3. Device



Oscilloscope

Current probe



Device

F. Measure Current

F1. Setup

Step 1: Use a current measuring probe clamped to the positive power supply wire (clamp in the direction of the current)

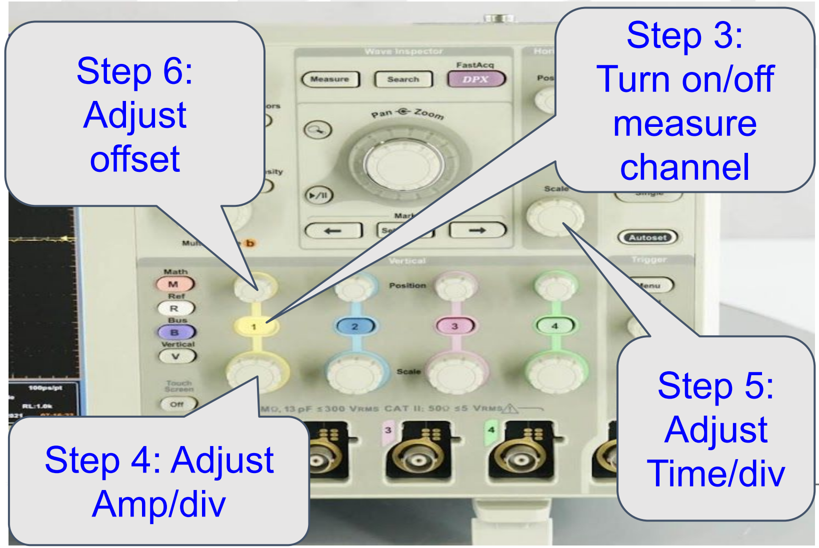
Step 2: Power on device need to measure

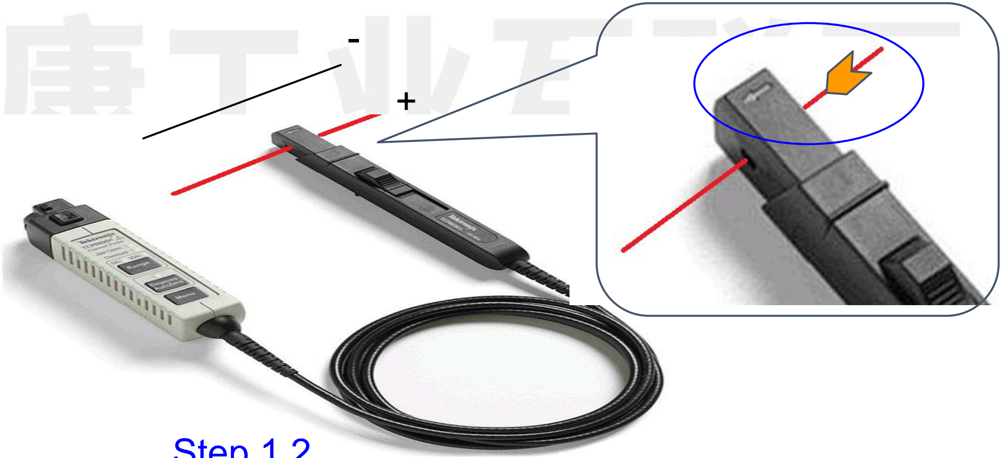
Step 3: Turn on channel to measure (ex: Channel 1 by press button “1” on the Oscilloscope)

Step 4: Adjust Amp/div by knob as bellow picture. Amp/div is to adjust the size of the Ampe scale

Step 5: Adjust Time/div by knob as bellow picture. Time/Div is to adjust the length of time scale

Step 6 (option): Adjust offset by knob as bellow picture. This function is to adjust the waveform displayed in the frame of the measurement screen. This function will be use if the waveform scale is so big.





Step 1,2

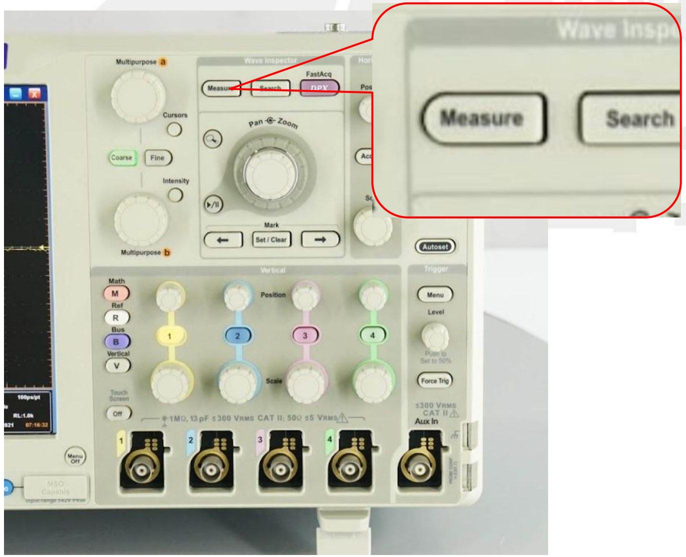
F1. Setup

Step 7: On the machine press “Measure” to show the “Measurement Setup” table

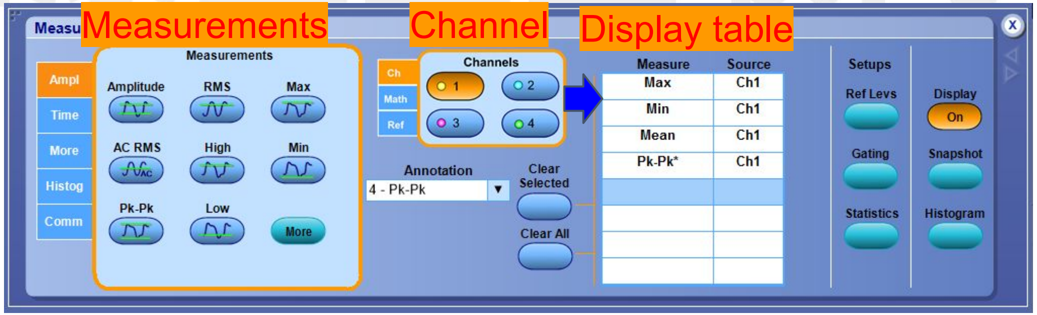
- + Select the “Channel” will be used to measure
- + Select the item to use in the “Measurements” table

And then the item that select before will be show on the “Display table”

Ex: Select Max, Min, Mean, Pk-Pk. These data will show directly on the measurement screen as below.



		Value	Mean	Min	Max	St Dev	Count	Info
C1	Max	34.56mA	42.960001m	34.56m	51.36m	11.88m	2.0	
C1	Min	30.48mA	31.080001m	30.48m	31.68m	848.5µ	2.0	
C1	Mean	32.84mA	37.046513m	32.84m	41.25m	5.942m	2.0	
C1	Pk-Pk	4.08mA	11.88m	4.08m	19.68m	11.03m	2.0	

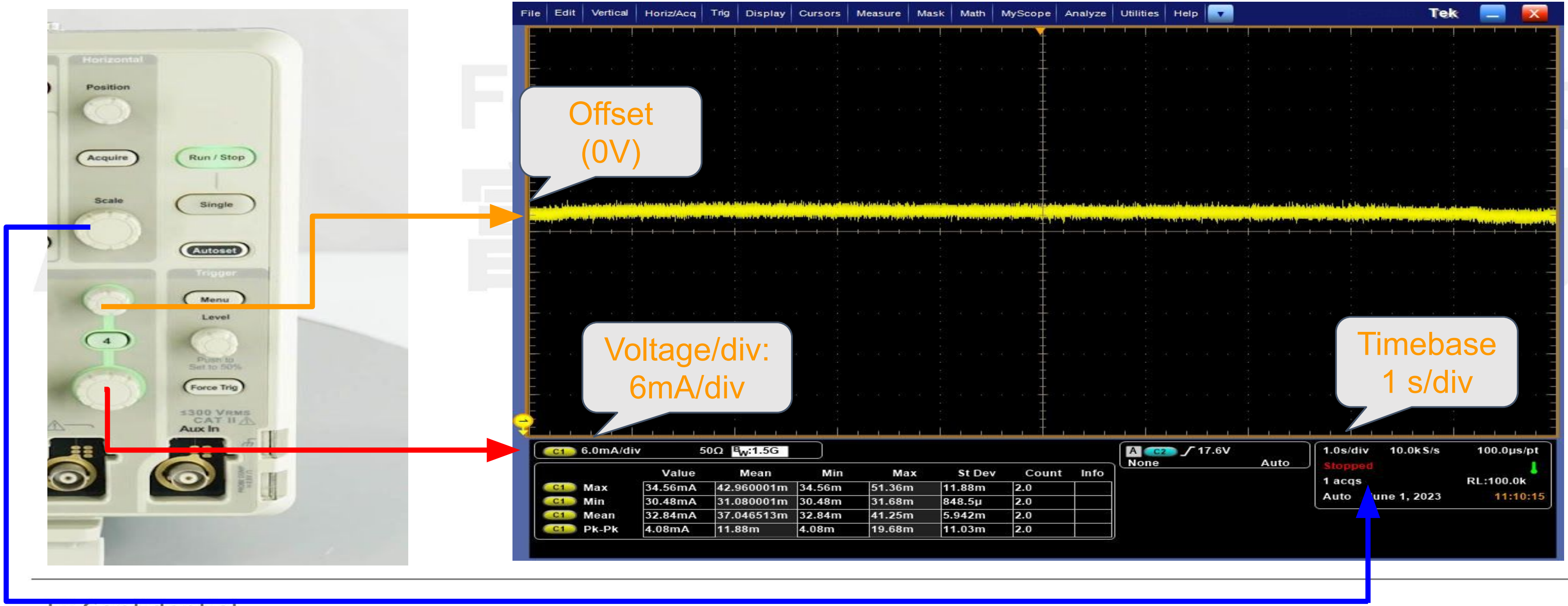


F2. Measure

Measure Current of 12V power supply

- Select Amp/div suitable with Voltage measured (ex: Select 6mA/div for measure small current)
- Select Timebase: If wants to see details chose Timebase small ex 10us/div

  If wants to see the overview chose Timebase large ex 1s/div



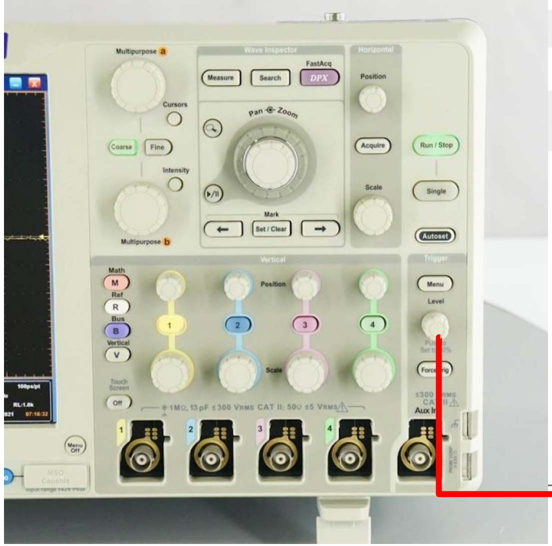
G. Measure Inrush Current

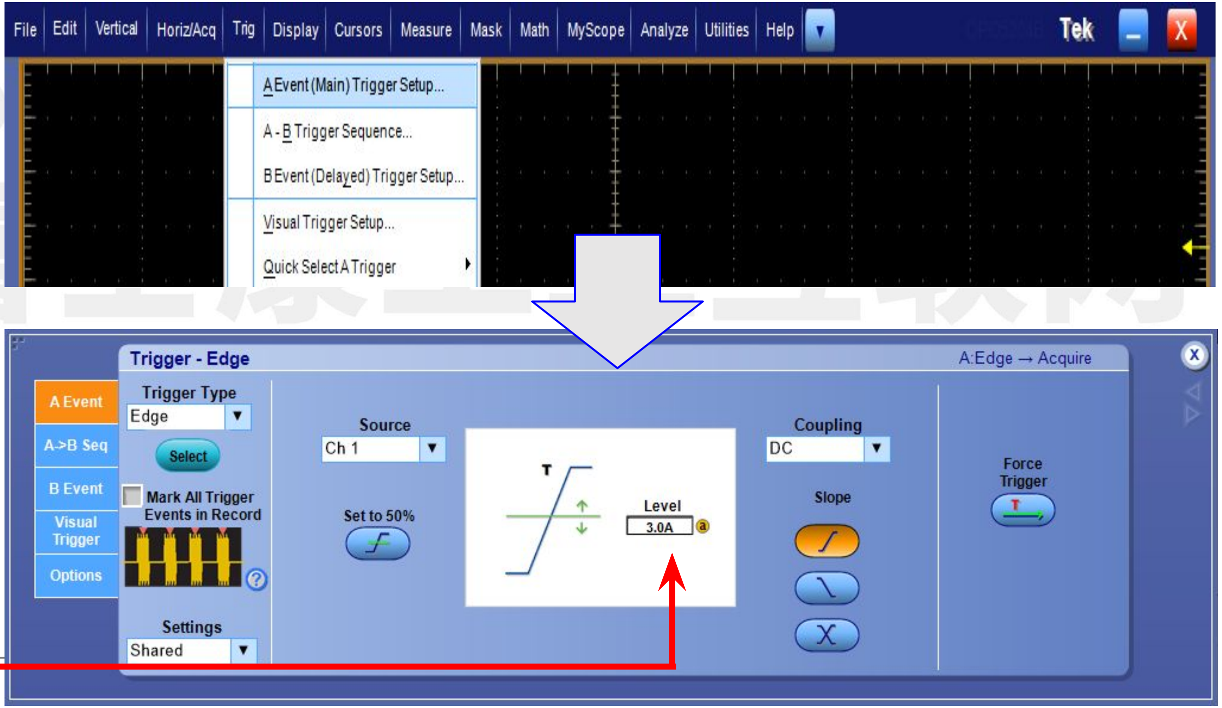
G1. Measure

Step 1 →7: Setup same measure current as above
Step 8: Setup the trigger

In screen select Trig / A Event (Main) Trigger Setup. → Will show Trigger-Edge table:
+ In Source select channel measure
+ In Trigger Type select Edge
+ In Slope select Rising edge pulse

Setup trigger about higher than the current normal usage level





G2. Measure

Step 9: In oscilloscope choose “Single”

Step 10: Power on the DUT and then adjust Vol/div and time/div accordingly